Kubernetes Deployment Assignment

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Batch 03 - Assignment 3 - Kubernetes

Introduction:

This assignment demonstrates deployment and management of an application using Kubernetes. The project includes containerization using Docker, deployment using Kubernetes resources, rolling updates, and health monitoring using probes.

Objective:

The main objective of this assignment is to understand:

- Pod creation
- ReplicaSet management
- Deployment management
- Rolling updates
- Kubernetes Services
- Container health checks using probes

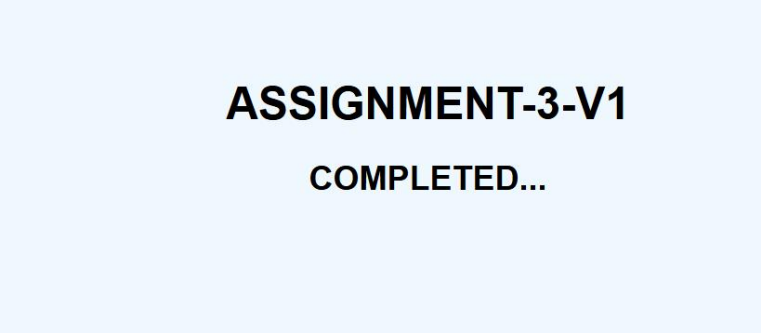
Frontend Application:

The application is a simple frontend UI created using HTML and CSS.

Version 1

Displays:

ASSIGNMENT-3-V1
COMPLETED…



Version 2

Displays:

ASSIGNMENT-3-V2
COMPLETED…

ASSIGNMENT-3-V2

COMPLETED...

Docker Setup:

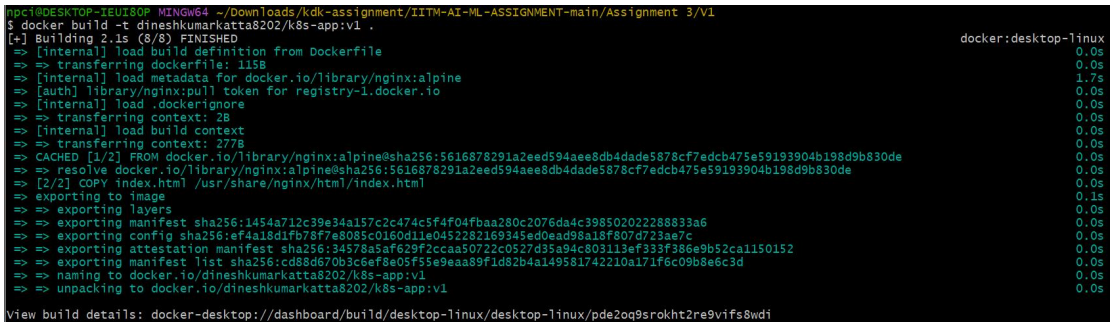
Dockerfile:

FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80

Docker Build Commands:

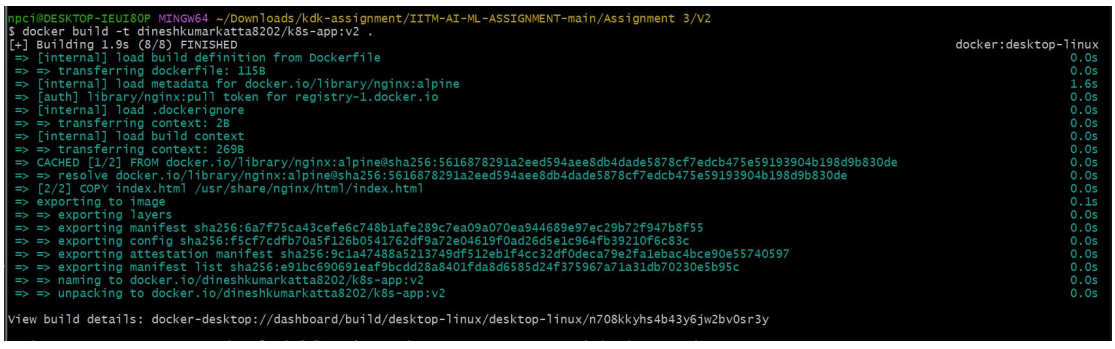
Build v1

docker build -t dineshkumarkatta8202/k8s-app:v1 .



Build v2

docker build -t dineshkumarkatta8202/k8s-app:v2 .



Docker Push Commands

docker push dineshkumarkatta8202/k8s-app:v1

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/V2
$ docker push dineshkumarkatta8202/k8s-app:v1
The push refers to repository [docker.io/dineshkumarkatta8202/k8s-app]
583599bb7d38: Layer already exists
4a8b0b2a5b19: Layer already exists
612c0c1df4c5: Layer already exists
aee4e54b3865: Layer already exists
b53c21b444f1: Already exists
6a0ac1617861: Layer already exists
82736a35d0e7: Layer already exists
6a76e7b4ac7c: Layer already exists
781ff50d2644: Layer already exists
453da7dbc73e: Layer already exists
v1: digest: sha256:cd88d670b3c6ef8e05f55e9eaa89f1d82b4a149581742210a171f6c09b8e6c3d size: 856
```

docker push dineshkumarkatta8202/k8s-app:v2

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/V2
$ docker push dineshkumarkatta8202/k8s-app:v2
The push refers to repository [docker.io/dineshkumarkatta8202/k8s-app]
dde76d73b798: Pushed
781ff50d2644: Layer already exists
6a0ac1617861: Layer already exists
82736a35d0e7: Layer already exists
4a8b0b2a5b19: Layer already exists
3503c548a3ce: Pushed
583599bb7d38: Layer already exists
aee4e54b3865: Layer already exists
453da7dbc73e: Layer already exists
612c0c1df4c5: Layer already exists
v2: digest: sha256:e91bc690691eaf9bcdd28a8401fda8d6585d24f375967a71a31db70230e5b95c size: 856
```

Kubernetes Files

pod.yml:

- Creates a single pod
- Runs container using v1 image

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl apply -f pod.yaml
pod/frontend-pod created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get pods
NAME           READY   STATUS    RESTARTS   AGE
frontend-pod   1/1     Running   0          11s
```

ReplicaSet (rs.yml):

- Maintains multiple pod replicas
- Ensures high availability 0

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
frontend-rs   3         3         3       65s
```

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl apply -f rs.yaml
replicaset.apps/frontend-rs created

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
frontend-pod        1/1     Running   0          3m46s
frontend-rs-bwddd   1/1     Running   0          7s
frontend-rs-c4fzk   1/1     Running   0          7s
```

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deployment.yml

- Manages application deployment
- Supports rolling updates
- Includes probes

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get deployments
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
frontend-deploy   3/3     3            3           11s

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
frontend-deploy-9bd7b7d6d-rxql8   1/1     Running   0          14s
frontend-deploy-9bd7b7d6d-s6rrt   1/1     Running   0          14s
frontend-deploy-9bd7b7d6d-st47r   1/1     Running   0          14s
frontend-pod                      1/1     Running   0          5m37s
frontend-rs-bwddd                 1/1     Running   0          118s
frontend-rs-c4fzk                 1/1     Running   0          118s
```

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service.yml

- Exposes application externally
- Routes traffic to pods

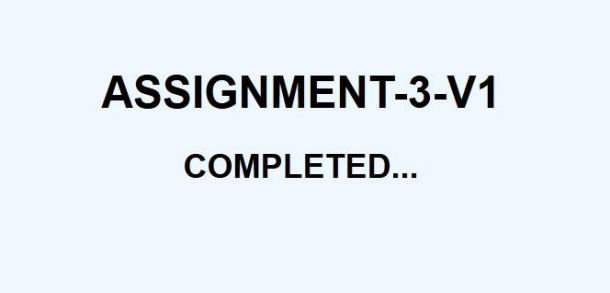
```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get svc
NAME               TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
frontend-service   NodePort    10.96.45.65    <none>        80:30080/TCP     10s
kubernetes         ClusterIP   10.96.0.1      <none>        443/TCP          123m
```

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl port-forward svc/frontend-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```

V1 Running:

Browser showing:

ASSIGNMENT-3-V1
COMPLETED…



Update Deployment to v2:

Run image update command:

$ kubectl set image deployment/frontend-deploy frontend=dineshkumarkatta8202/k8s-app:v2

This command updates deployment image from:

- v1 → v2

Monitor Rolling Update

Watch update process:

kubectl get pods -w

During update:

Old Pods terminate gradually

New Pods create gradually

Example process:

Old Pod → Terminating
New Pod → Creating
New Pod → Running

```
npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl set image deployment/frontend-deploy frontend=dineshkumarkatta8202/k8s-app:v2
deployment.apps/frontend-deploy image updated

npci@DESKTOP-IEUI8OP MINGW64 ~/Downloads/kdk-assignment/IITM-AI-ML-ASSIGNMENT-main/Assignment 3/k8s
$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
frontend-deploy-7b66c7ff95-9jkxx  1/1     Running   0          55s
frontend-deploy-7b66c7ff95-h4hv9  1/1     Running   0          50s
frontend-deploy-7b66c7ff95-vsk4b  1/1     Running   0          55s
frontend-pod                      1/1     Running   0          14m
frontend-rs-bwddd                 1/1     Running   0          10m
frontend-rs-c4fzk                 1/1     Running   0          10m
```

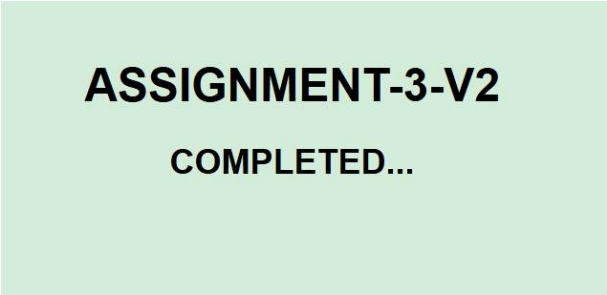
---

v2 Running

Browser showing:

ASSIGNMENT-3-V2

COMPLETED…



Conclusion:

This assignment successfully showcased the process of deploying and managing an application using Kubernetes in a practical environment.

The frontend application was first containerized with Docker and then deployed to a Kubernetes cluster using core Kubernetes components such as Pods, ReplicaSets, Deployments, and Services. To ensure the application remained stable and reliable, health checks were configured using Liveness, Readiness, and Startup Probes.

Initially, Version 1 of the application was deployed, and later it was upgraded to Version 2 through the Rolling Update strategy. During this process, Kubernetes smoothly replaced the old Pods with new ones without causing any downtime, ensuring uninterrupted application availability.

Through this assignment, valuable hands-on experience was gained in Docker image creation, version management, Kubernetes resource handling, service exposure, rolling updates, and container health monitoring.

Overall, the assignment provided a clear understanding of the Kubernetes deployment lifecycle and demonstrated how applications can be efficiently managed and updated in a containerized environment.